Department of Computer Science

Neural Network based emulation of SGS turbulence in MONC: comparing physics-guided multi-task learning against data-driven closure in online-coupled LES

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Central research questions:

- **Can we train a Neural Network (NN) based emulator, "offline", for the subgrid-scale (SGS) turbulence parameterization inside Met Office NERC Cloud Model (MONC)?**
- **Will this NN based SGS emulator produce stable and realistic simulations across different resolutions and atmospheric regimes?**

Result: Yes, we can! The extensive evaluation show the NN SGS closure realistic updraft anatomy and realistic climate-change response over +2-4 K warming scenarios and a 2× grid resolution change (pattern r ≥ 0.97).

Trained on sparsely sampled data, deployed using FTorch coupling

- **Only ~5% of the data volume used:** 4.05 M regime-stratified point samples, sparsely drawn from a 83.9 M mixed-resolution archive — RCE deep convection (Δx = Δy = 1 km) + ARM shallow cumulus (Δx = Δy = 50 m).
- **One multi-task network,** 54-feature stencil inputs → point-wise, grid-agnostic K_m, K_h (+ Ri & regime as auxiliary tasks); forward-pass Ri-conditioning; uncertainty task-weighting — winner of a 54-configuration ablation (WCCI-IJCNN & PASC 2026).
- **Two emulators, one recipe:** Ri-MLP (physics-guided, ≈442k params) vs Q4-MLP (data-driven ablation, ≈359k params), coupled in-process into MONC via FTorch (TorchScript → libtorch; no Python at runtime).
- **Four online legs vs static Smagorinsky:** RCE based simulations with CTRL (1 km), +2 K and +4 K SST warming (≈105–420 h records), and an unseen 500 m grid.

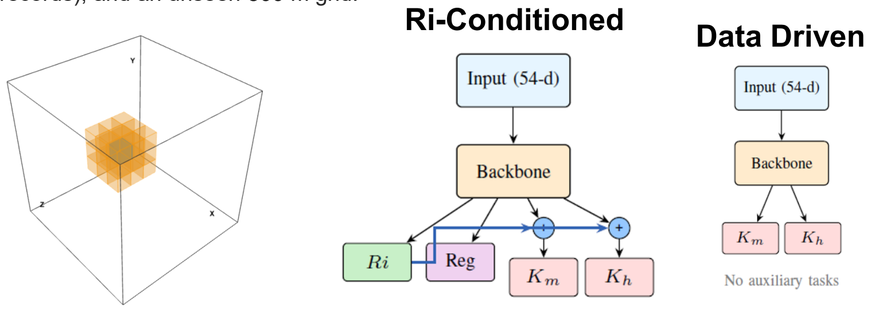
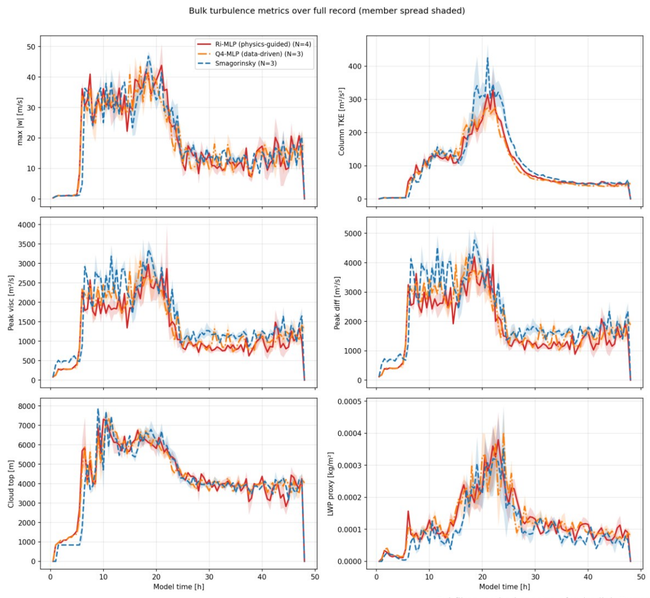
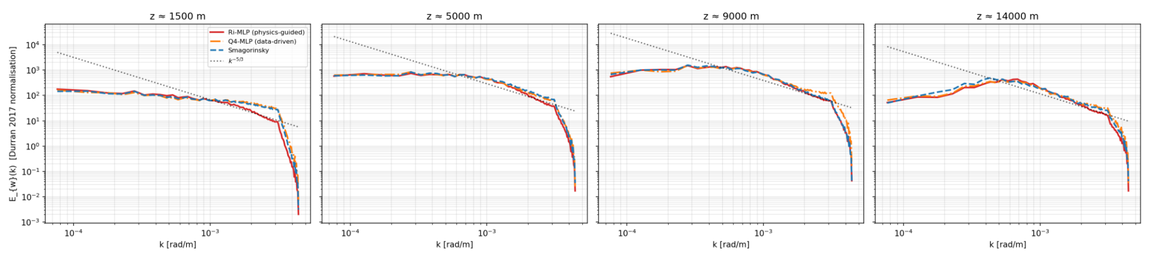


Fig. 1: Sparse star-stencil point sampling from 3-D LES fields (left) and the two deployed architectures (right): Ri-MLP — Ri predicted, then fed into the coefficient heads' forward pass; Q4-MLP — same backbone, direct coefficient regression, no auxiliary tasks.

1 · Online fidelity: 1 km (5 members) and 500 m (3 members) ensemble runs for RCE

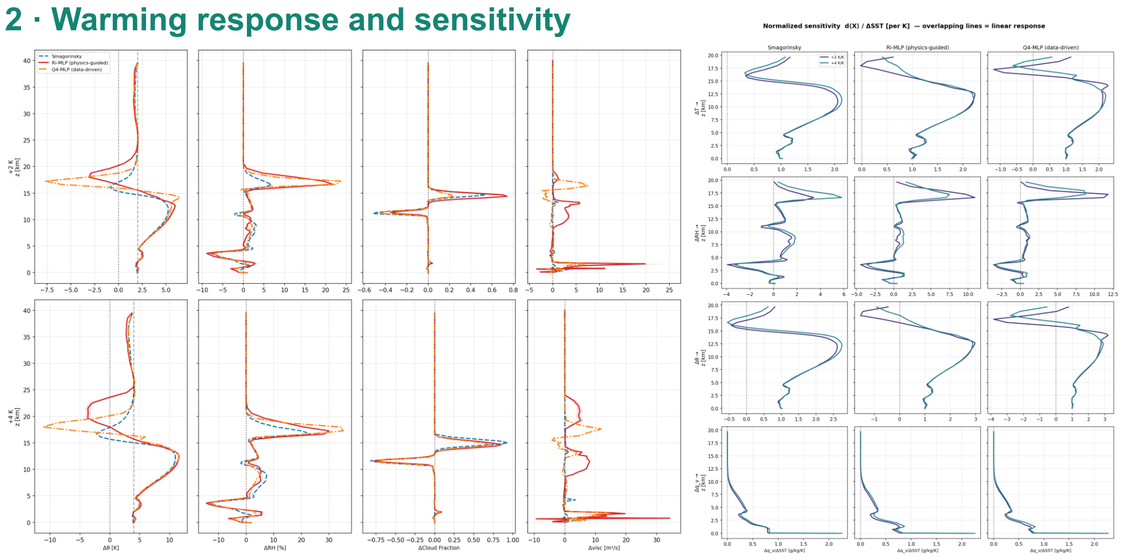


- **48-h ensembles at the trained grid (Fig. 2):** N = 5 members per scheme; both emulators track Smagorinsky's full convective life cycle.
- Max |w|, column TKE, peak Km, cloud-top & LWP follow Smagorinsky, with no drift or blow-up.
- **At the unseen 500 m grid** the state is statistically indistinguishable: KGE(θ) 0.91 / 0.93, $R^2(q_v) \approx 0.999$, domain-PDF KL ≤ 0.017.
- **Spectral fidelity at 500 m (Fig. 3):** subfilter w-spectra (Durran 2017; radial) reproduce Smagorinsky's spectral shape & near-grid damping at z ≈ 1.5–14 km.



Figs. 2–3: (Top panel)1 km ensemble bulk time series (spread shaded) and (bottom panel) 500 m subfilter-method w power spectra at z ≈ 1.5 / 5 / 9 / 14 km (500 m: single members, early window).

2 · Warming response and sensitivity

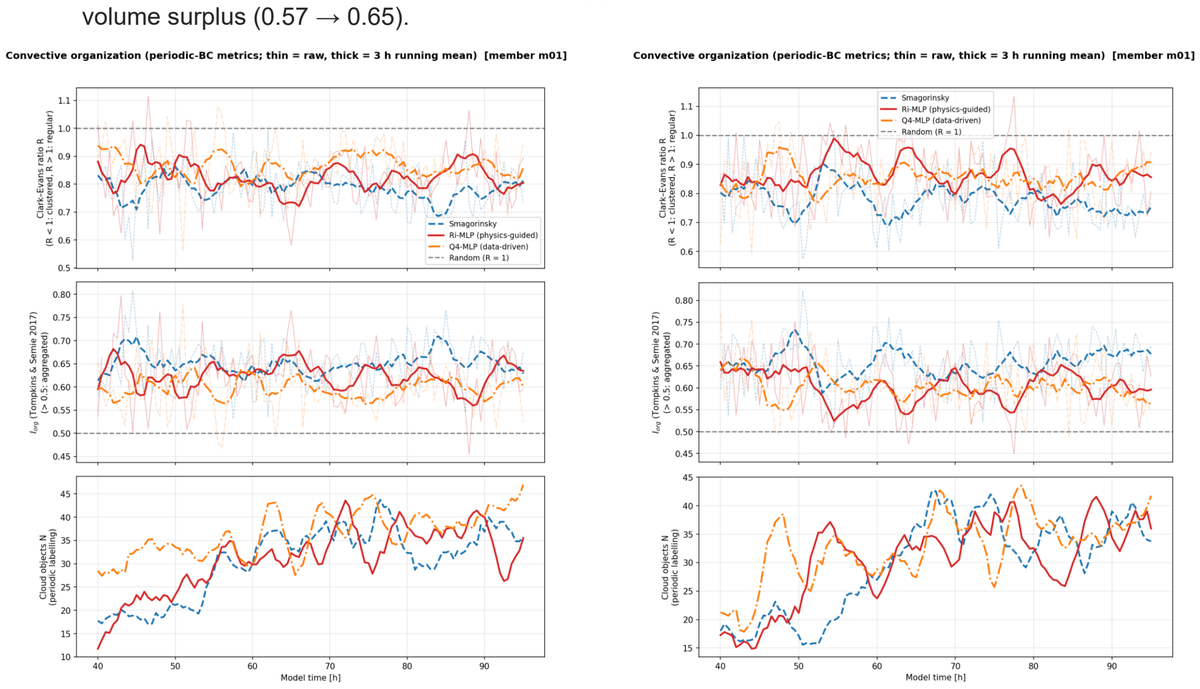


Figs. 4–5: (Left) Warming-response Δ = mean(warm) − mean(control) profiles at +2 K and +4 K (selected variables): all three closures agree through the troposphere; divergence confined to 15–20 km. (Right) Per-K normalized sensitivity: near-linear +2 K/K vs +4 K/K response.

3 · Convective organization signatures under warming scenarios

Convective organization (periodic Clark–Evans & I_org, Figs. 6–7: +2 K warming (left), +4 K warming (right)).

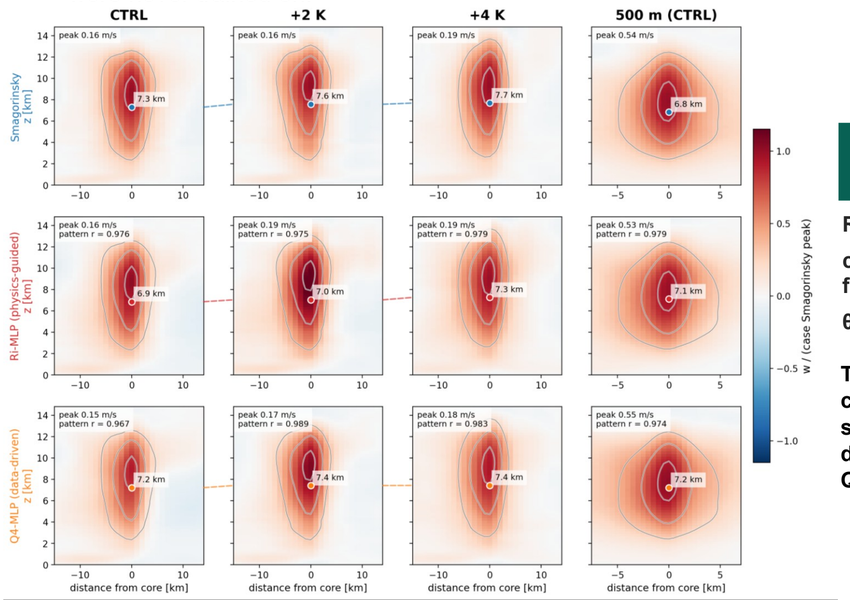
- ❑ Both learned closures produce more numerous, more dispersed convective cells than Smagorinsky.
- ❑ This is observed as a shared fingerprint, consistent across warming scenarios.
- ❑ They also exhibit earlier onset of convective triggering (coverage gap 1.2× → 4.7×) and a cloud-volume surplus (0.57 → 0.65).



Figs. 6–7: Organization metrics at +2 K and +4 K (member m01).

4 · Physical realism across distribution shifts and unseen regimes

ML SGS closures inside MONC reproduce the reference updraft across a resolution and climate they were never trained on.



KGE (0→4 K)	Ri-MLP	Q4-MLP
RH	0.98–0.99	0.98–0.99
cloud fraction	0.86–0.97	0.84–0.93
θ′	0.88–0.93	0.89–0.90

Table 1: State metrics maintain consistency across warming scenarios, with marginal difference between Ri-MLP and Q4-MLP

Fig. 8: Canonical updraft composites, 3 closures × {CTRL, +2 K, +4 K, 500 m}; white contours = Smagorinsky plume. Pattern r = 0.97–0.99 for both emulators at every rung. Core ascent ~0.1 km K⁻¹ reproduced by all 3 runs. (Smag 7.3→7.7 km); 500 m peaks 0.54 / 0.53 / 0.55 m s⁻¹. Three-way near-identical results.

5 · Warming response, sensitivity & fingerprints

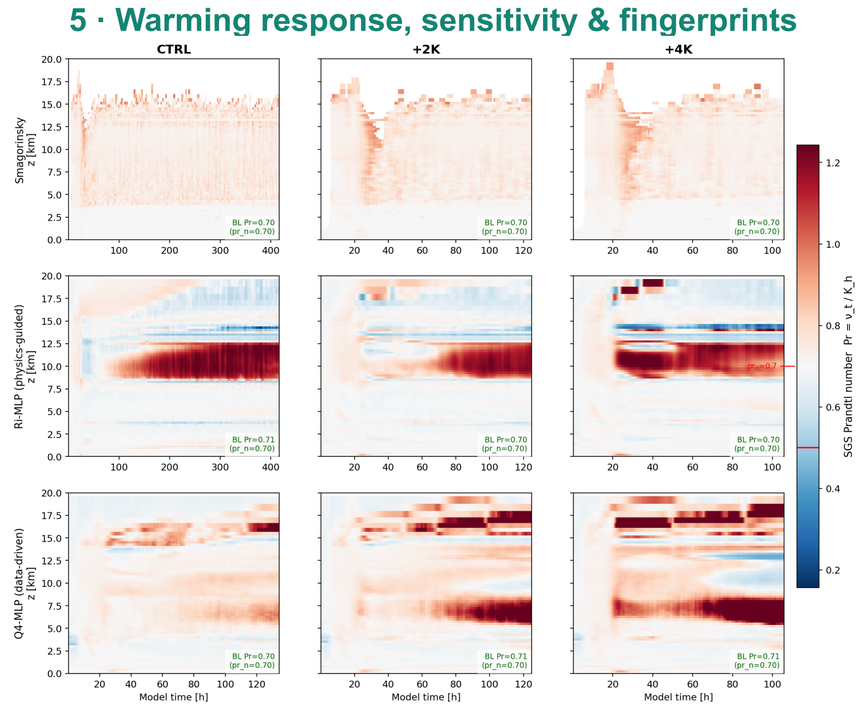


Fig 9: SGS Prandtl number Pr = ν_t/K_h, time–height per closure × climate (under warming scenarios) for RCE at 1 km horizontal resolution.

❖ SGS Prandtl number per closure: Smagorinsky (fixed f_m/f_h of Ri); ML (emergent property from independent K_m and K_h heads).

❖ Every scheme recovers the neutral-BL Pr = 0.70–0.71 (= MONC pr_n) — a physics validation for the independent K heads.

❖ Ri-MLP's elevated-Pr layer is anvil-anchored (10–13 km) and climate-consistent, while Q4-MLP's sits lower (5–8 km) and intensifies with warming.

References & data

1. Panda et al., IEEE WCCI-IJCNN 2026 (oral).
2. Panda et al., PASC 2026.
3. Durran et al., MWR 2017.

Acknowledgements & contact

Met Office (CASE studentship & HPC) · JASMIN · AFESP, Univ. of Reading · ICCS / FTorch, Cambridge.

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